CH10. Fundamentals of the Metal-Oxide-Semiconductor Field-Effect Transistor (Part II)

Prof. Sung Hun Jin
Department of Information Display,
Kyung Hee University, Republic of Korea

10.3 The Basic MOSFET Operation

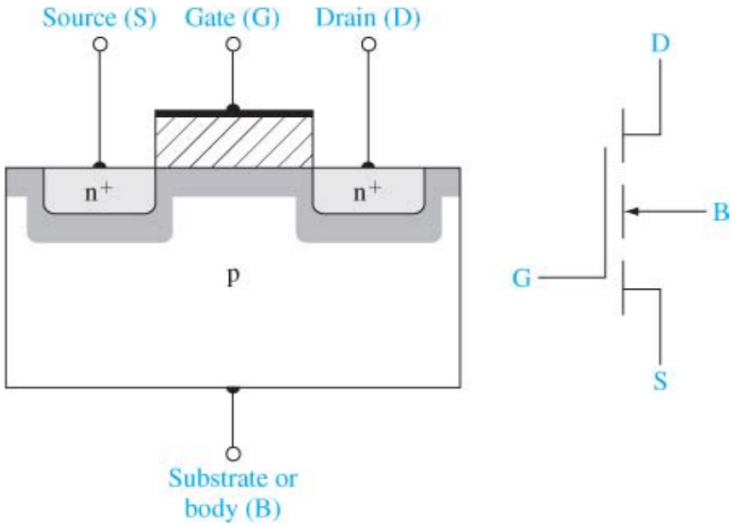


Figure 10.34 | Cross section and circuit symbol for an n-channel enhancement mode MOSFET.

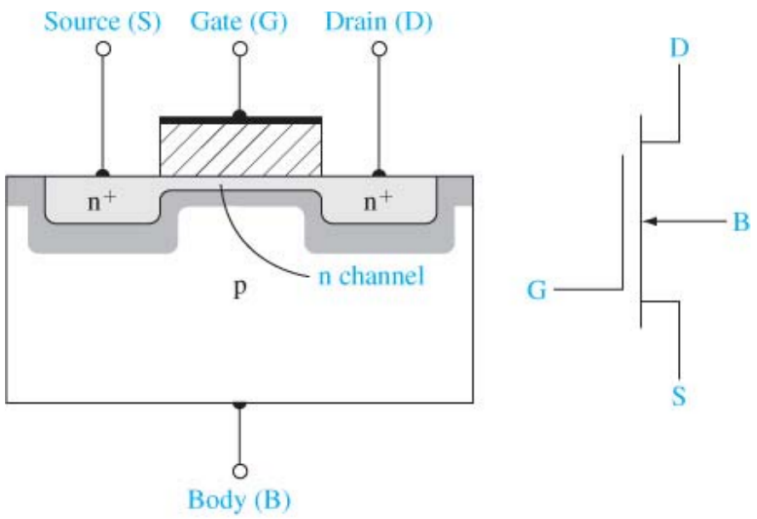


Figure 10.35 | Cross section and circuit symbol for an n-channel depletion mode MOSFET.

- The current in a MOSFET is due to the flow of charge;
  - ✓ Inversion layer
  - ✓ Channel region adjacent to the oxide-semiconductor interface.
- Inversion layer charge in enhancement-type MOS capacitors.
- Depletion type device (i.e., channel already exist at zero gate voltage)

10.3.1 MOSFET Structures

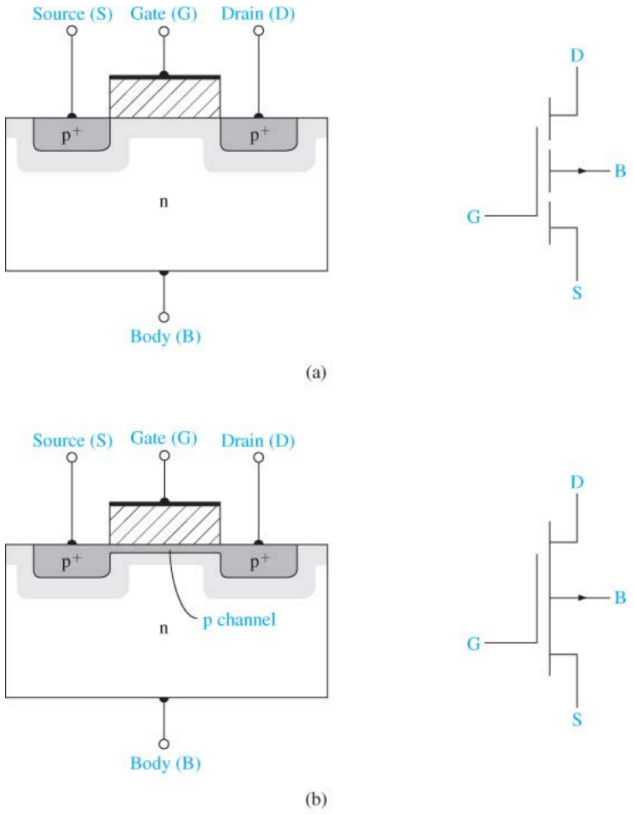


Figure 10.36 | Cross section and circuit symbol for (a) a p-channel enhancement mode MOSFET and (b) a p-channel depletion mode MOSFET.

- P-channel device operation
  - ✓ Substrate is (?) type.
- N-channel device operation.
  - ✓ Substrate is (?) type.

10.3.2 Current-Voltage Relationship-concepts

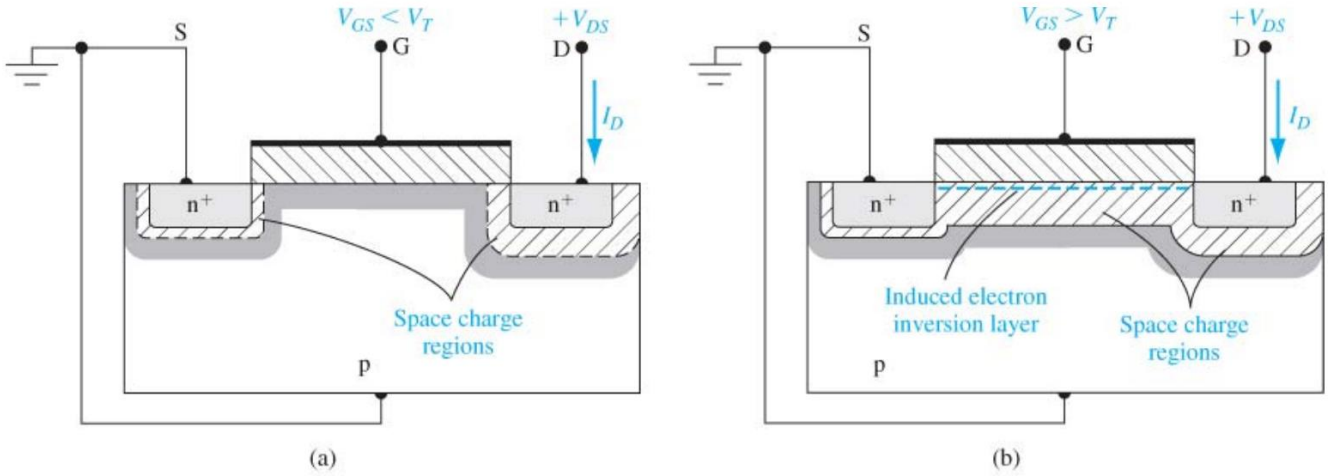


Figure 10.37 | The n-channel enhancement mode MOSFET (a) with an applied gate voltage $V_{GS} < V_T$ and (b) with an applied gate voltage $V_{GS} > V_T$.

- Fig. 10.37 (a)
  - ✓ N-channel enhancement mode MOSFET with a gate-to-source voltage.
  - ✓ $V_{GS}<V_{TH}$ and V_{DS}~ very small.
  - ✓ Source and body ➔ ground potential.
  - ✓ No inversion layer
  - ✓ Drain-to-substrate pn junction is reverse biased. ➔ ID current is zero.

10.3.2 Current-Voltage Relationship-concepts

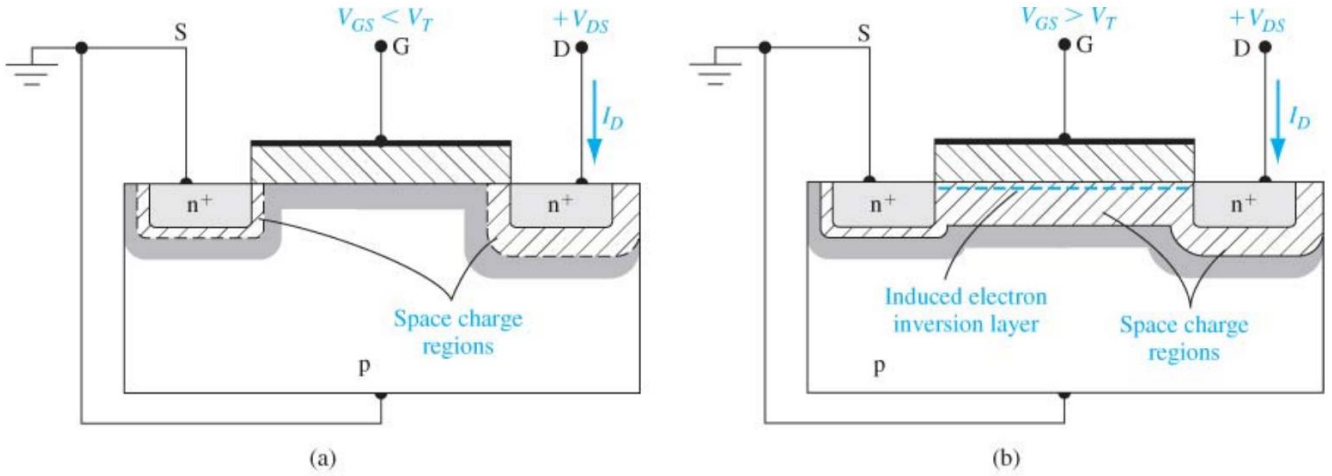


Figure 10.37 | The n-channel enhancement mode MOSFET (a) with an applied gate voltage $V_{GS} < V_T$ and (b) with an applied gate voltage $V_{GS} > V_T$.

- Fig. 10.37 (b)
  - ✓ N-channel enhancement mode MOSFET with a gate-to-source voltage.
  - ✓ $V_{GS} > V_{TH}$ ➔ inversion layer has been created.
  - ✓ V_{DS}~ very small ➔ Lateral electric field
  - ✓ Electrons will flow from source to drain.

10.3.2 Current-Voltage Relationship-concepts

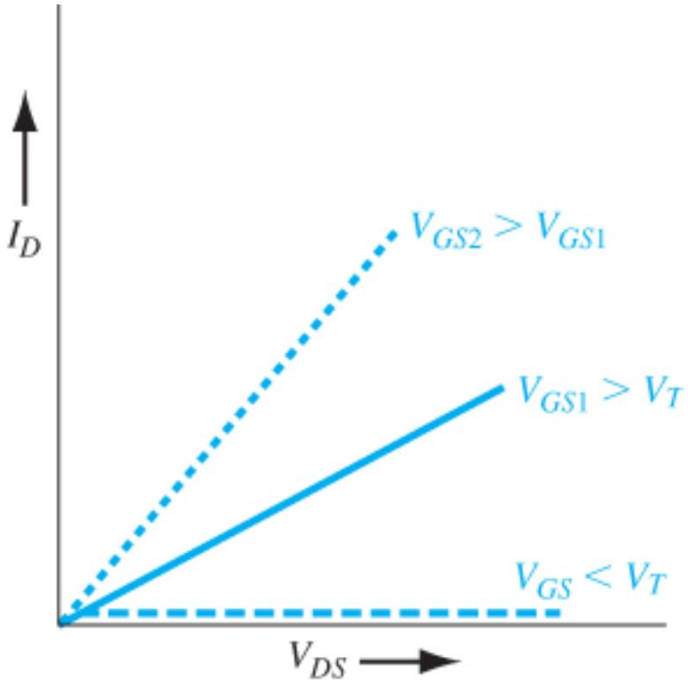


Figure 10.38 | I_D versus V_{DS} characteristics for small values of V_{DS} at three V_{GS} voltages.

- Small V_{DS}, $V_{GS}>V_{TH}$
  - ✓ Channel region has the characteristics of a resistor.

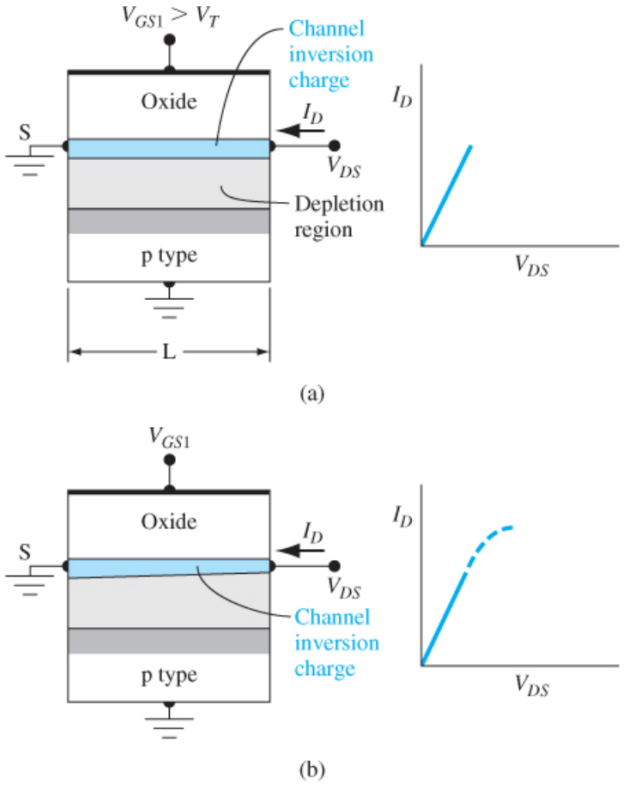
$$I_D = g_d V_{DS} \quad (10.41)$$

  - ✓ G_d is defined as the channel conductance in the limit as $V_{DS}\rightarrow 0$

$$g_d = \frac{\partial I_{DS}}{\partial V_{DS}} = \frac{W}{L} \cdot \mu_n \left| Q'_n \right| \quad (10.42)$$

  - ✓ Where μn is mobility of electrons.
  - ✓ Q_n': inversion layer charge is a function of the gate voltage.
- The basic MOSFET action
  - ✓ Modulation of channel conductance.
  - ✓ By V_{GS} (gate bias.)

10.3.2 Current-Voltage Relationship-concepts



- Fig. 10.39 (a)
  - ✓ $V_{GS}>V_{TH}$, V_{DS}~very small.
  - ✓ Channel inversion charge
  - ✓ At small V_{DS}, channel conductance.
- Fig. 10.39 (b)
  - ✓ As V_{DS} increases, voltage drop across oxide near drain terminal decrease➔ inversion charge decrease➔ g_d decrease➔ slope change in I_{DS}-V_{DS}

10.3.2 Current-Voltage Relationship-concepts

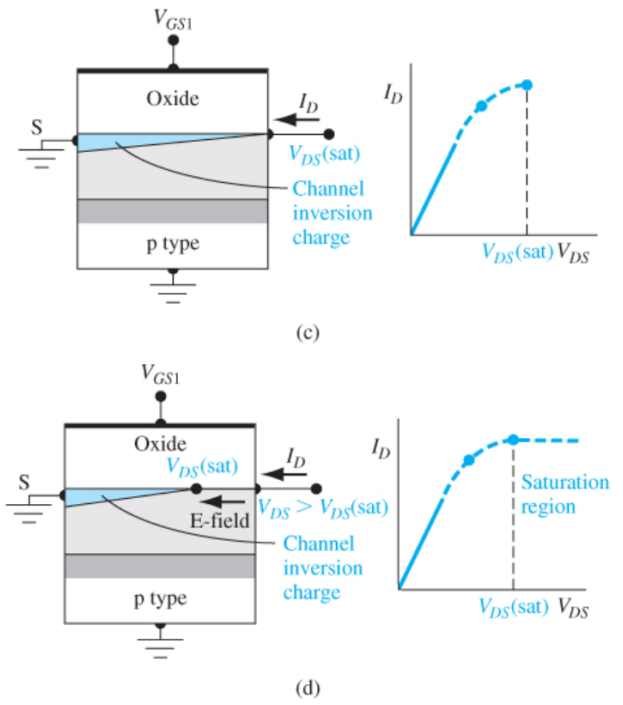


Figure 10.39 | Cross section and I_D versus V_{DS} curve when $V_{GS} < V_T$ for (a) a small V_{DS} value, (b) a larger V_{DS} value, (c) a value of $V_{DS} = V_{DS}$(sat), and (d) a value of $V_{DS} > V_{DS}$(sat).

- Fig. 10.39 (c)
  - ✓ As V_{DS} increases upto;.
  - ✓ V_{GS}-V_{TH}=V_{DS}(sat)
  - ✓ Incremental channel conductance of g_d~ 0
  - ✓ Slope=0 in I_{DS}-V_{DS}.
  - ✓ V_{DS}(sat) producing zero-inversion charge density at the drain terminal.
- Fig. 10.39 (d)
  - ✓ When V_{DS} become larger than V_{DS}(sat), the point in inversion charge is just zero➔ moves toward the source terminal.
  - ✓ Electrons are injected into the space charge region.
  - ✓ Why?
  - ✓ Saturation region.

10.3.2 Current-Voltage Relationship-concepts

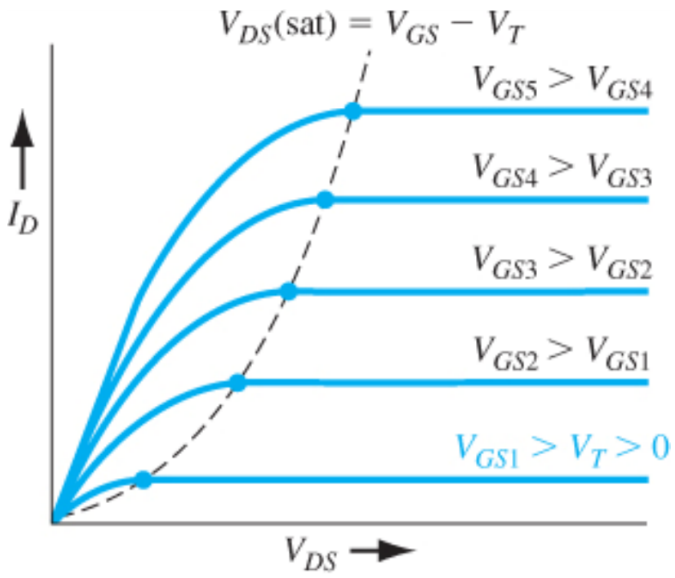


Figure 10.40 | Family of I_D versus V_{DS} curves for an n-channel enhancement mode MOSFET.

- Fig. 10.39 (c)
  - ✓ As V_{DS} increases upto;.
  - ✓ V_{GS}-V_{TH}=V_{DS}(sat)
  - ✓ Incremental channel conductance of g_d~ 0
  - ✓ Slope=0 in I_{DS}-V_{DS}.
  - ✓ V_{DS}(sat) producing zero-inversion charge density at the drain terminal.
- Fig. 10.39 (d)
  - ✓ When V_{DS} become larger than V_{DS}(sat), the point in inversion charge is just zero➔ moves toward the source terminal.
  - ✓ Electrons are injected into the space charge region.
  - ✓ Why?
  - ✓ Saturation region.

10.3.2 Current-Voltage Relationship-concepts

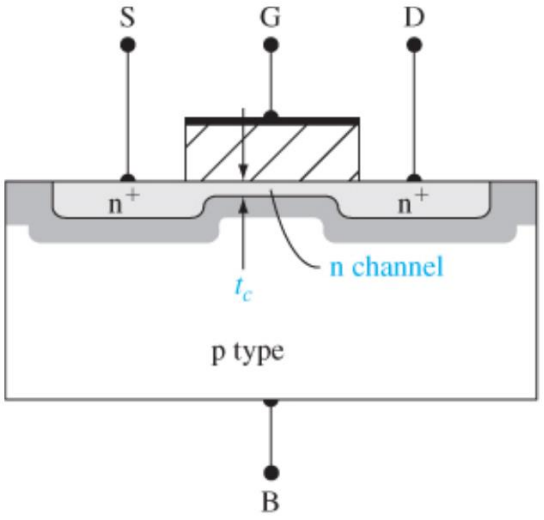


Figure 10.41 | Cross section of an n-channel depletion mode MOSFET.

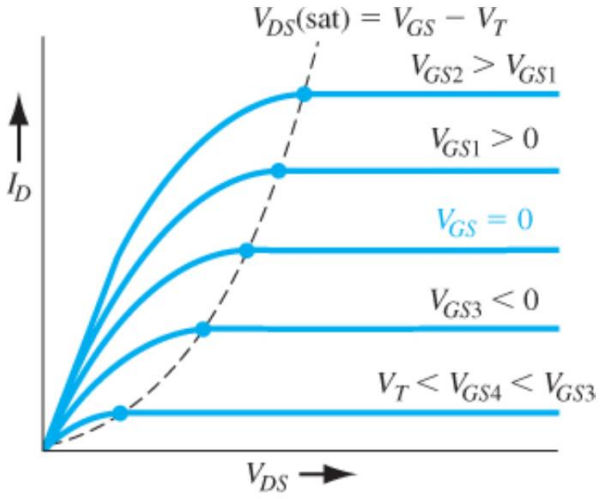


Figure 10.42 | Family of I_D versus V_{DS} curves for an n-channel depletion mode MOSFET.

- Depletion mode operation
  - ✓ N-channel region is actually induced electron inversion charge
  - ✓ Metal work function difference
  - ✓ Fixed charge in the oxide
  - ✓ I-V characteristics is exactly the same. (only difference is V_{TH}<0V)

10.3.2 Derive the ideal current-voltage relation

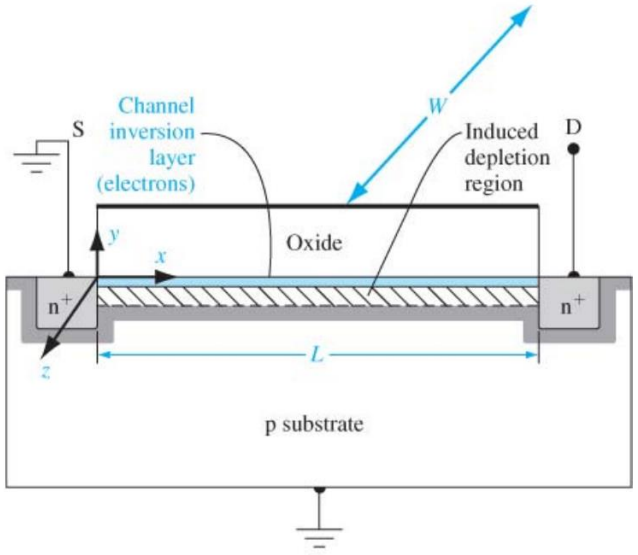


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

$$I_D = \frac{W\mu_n C_{ox}}{2L}\left[2(V_{GS}-V_T)V_{DS}-V_{DS}^2\right] \quad (10.44a)$$

Where can be written as (10.44b) or (10.44c)

$$I_D = \frac{k'_n}{2}\cdot\frac{W}{L}\cdot\left[2(V_{GS}-V_T)V_{DS}-V_{DS}^2\right] \quad (10.44b)$$

$$I_D = K_n\cdot\left[2(V_{GS}-V_T)V_{DS}-V_{DS}^2\right] \quad (10.44c)$$

- $K_n'=\mu_n C_{ox}$:process conduction parameter for the n-channel MOSFET (unit: A/V^2)
- $K_n=(W\ \mu_n C_{ox})/2L=(k_n'/2)*(W/L)$ is called " the conduction parameter" for the n channel MOSFET.

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

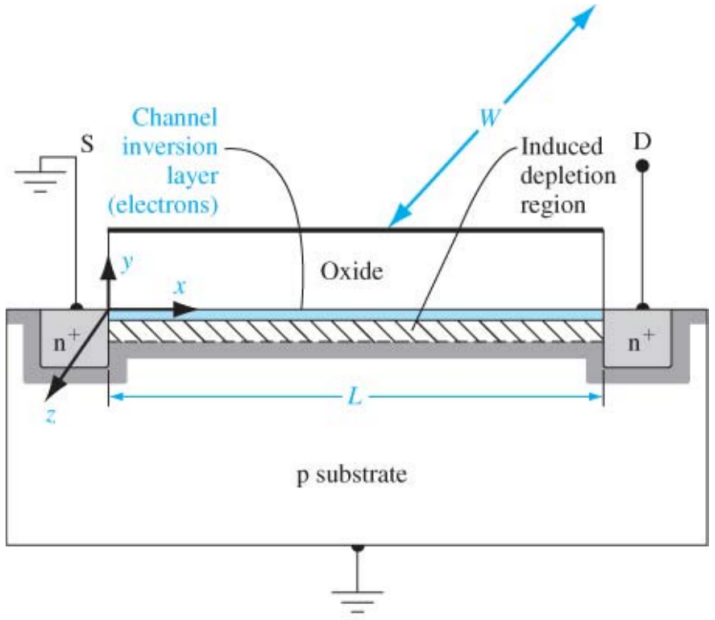


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

1. The current in the channel is due to drift rather than diffusion.
2. There is no current through the gate oxide.
3. A gradual channel approximation is used in which $\frac{\partial E_y}{\partial y} >> \frac{\partial E_x}{\partial x}$
4. This approximation means that E_x is essentially a constant.
5. Any fixed oxide charge is an equivalent charge density at the oxide –semiconductor interface.
6. The carrier mobility in the channel is constant.

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

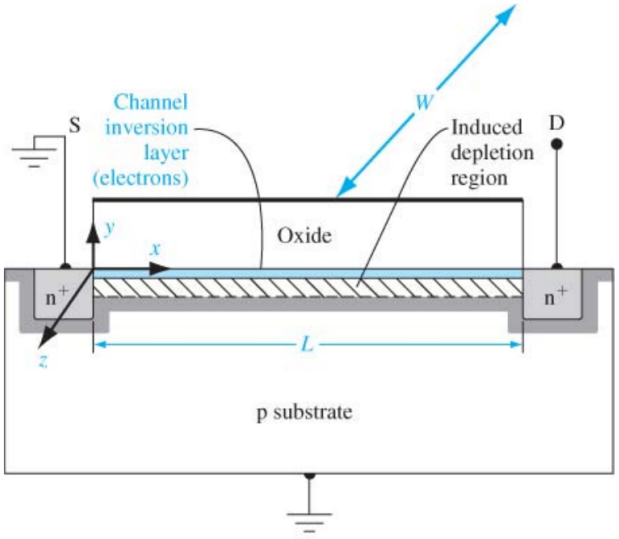


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

1. Start with Ohm's law:

$$J_x = \sigma E_x \quad (10.46)$$

2. Total current is found by integrating J_x over cross-sectional area in the y and z direction.

$$I_x = \int_y \int_z J_x dy dz \quad (10.47)$$

$$Q'_n = \int en(y) dy \quad (10.48)$$

: inversion layer charge per unit area.
W is the channel width

$$I_x = -W \mu_n Q'_n E_x \quad (10.49)$$

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

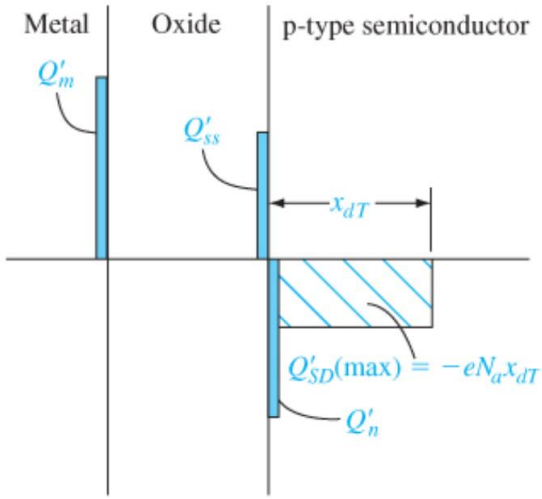


Figure 10.44 | Charge distribution in the n-channel enhancement mode MOSFET for $V_{GS} > V_T$.

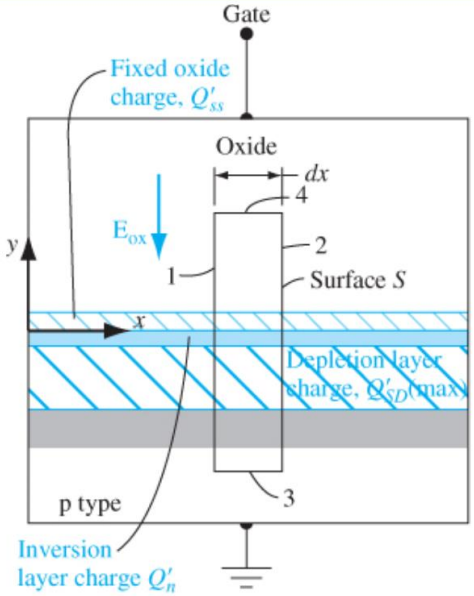


Figure 10.45 | Geometry for applying Gauss's law.

$$Q'_m + Q'_{ss} + Q'_n + Q'_{SD}\ (\max) = 0 \quad (10.50)$$

$$Gauss's\ law: \oint_S \varepsilon E_n dS = Q_T \quad (10.51)$$

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

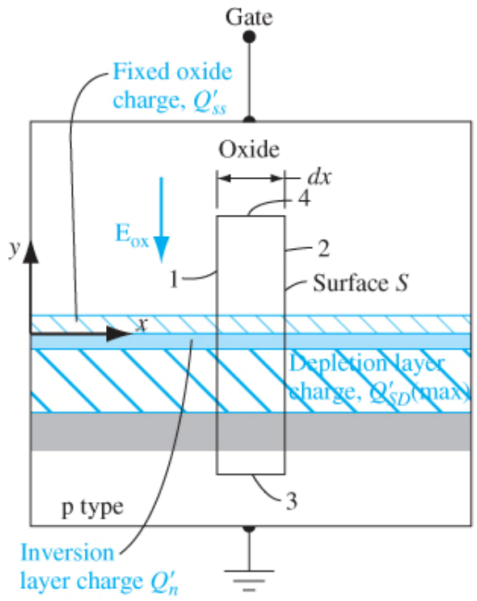


Figure 10.45 | Geometry for applying Gauss's law.

$$\oint_S \varepsilon E_n dS = -\varepsilon_{ox} E_{ox} W dx = Q_T \quad (10.52)$$

$$Q_T = [Q'_{ss} + Q'_n + Q'_{SD}(\max)] W dx \quad (10.53)$$

$$-\varepsilon_{ox} E_{ox} = Q'_{ss} + Q'_n + Q'_{SD}(\max) \quad (10.54)$$

$$E_{Fp} - E_{Fm} = e(V_{GS} - V_x) \quad (10.55)$$

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

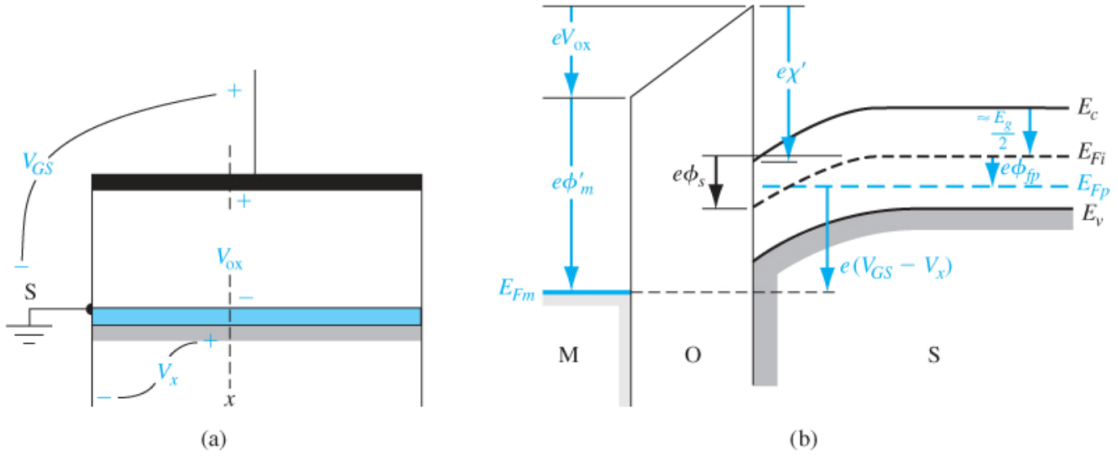


Figure 10.46 | (a) Potentials at a point x along the channel. (b) Energy-band diagram through the MOS structure at the point x.

$$E_{Fp} - E_{Fm} = e(V_{GS} - V_x) \quad (10.55)$$

Considering the potential barriers,

$$V_{GS} - V_x = (\phi'_m + V_{ox}) - \left(\chi' + \frac{E_g}{2e} - \phi_s + \phi_{fp} \right) \quad (10.56)$$

which can be also written as.

$$V_{GS} - V_x = V_{ox} + 2\phi_{fp} + \phi_{ms} \quad (10.57)$$

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

$$E_{ox} = \frac{V_{ox}}{t_{ox}} \quad (10.58)$$

$$-\varepsilon_{ox} E_{ox} = \frac{-\varepsilon_{ox}}{t_{ox}}\left[(V_{GS} - V_x) - (\phi_{ms} + 2\phi_{fp})\right] = Q'_{ss} + Q'_n + Q'_{SD}\,(\max) \quad (10.59)$$

$$I_x = -W\mu_n C_{ox}\frac{dV_x}{dx}\left[(V_{GS} - V_x) - V_T\right] \quad (10.60)$$

$$\int_0^L I_x dx = -W\mu_n C_{ox}\int_{V_x(0)}^{V_x(L)} \left[(V_{GS} - V_T) - V_x\right] dV_x \quad (10.61)$$

$$I_D = \frac{W\mu_n C_{ox}}{2L}\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right] \quad (10.62)$$

10.3.3 Current-Voltage Relationship -Mathematical Derivation.

$$I_D = \frac{k_n'}{2}\cdot\frac{W}{L}\cdot\left[2(V_{GS}-V_T)V_{DS}-V_{DS}^2\right] = K_n\left[2(V_{GS}-V_T)V_{DS}-V_{DS}^2\right] \quad (10.63)$$

$$V_{DS} = V_{GS} - V_T \quad (10.64)$$

$$I_D(sat) = \frac{W\mu_n C_{ox}}{2L}\left[2(V_{GS}-V_T)V_{DS}(sat)-V_{DS}^2(sat)\right] \quad (10.65)$$

$$I_D(sat) = \frac{W\mu_n C_{ox}}{2L}(V_{GS}-V_T)^2 \quad (10.66)$$

$$I_D = \frac{k_n'}{2}\cdot\frac{W}{L}\cdot(V_{GS}-V_T)^2 = K_n(V_{GS}-V_T)^2 \quad (10.67)$$

$$I_D = \frac{W\mu_n C_{ox}}{L}(V_{GS}-V_T)V_{DS} \quad (10.68)$$

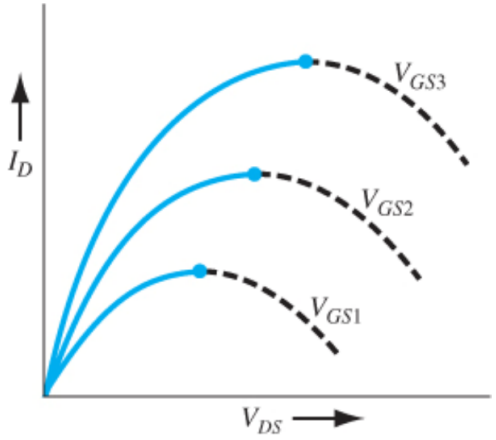


Figure 10.47 | Plots of I_D versus V_{DS} from Equation (10.62).

Example 10.7

① **Design the width of a MOSFET such that a specified current is induced for a given applied bias**

- **Assume the following transistor parameter : L=1.25μm, μ_n=650cm^2/V-s,**
- **C_{ox}=6.9× 10^{-8}F/cm^2, V_T=0.65V**
- **I_D(sat)=4mA for V_{GS}=5V**

For the transition biased in the saturation region, we have, from Equation (10.66),

$$I_D(\text{sat}) = \frac{W\mu_n C_{ox}}{2L}(V_{GS} - V_T)^2$$

or

$$4\times10^{-3} = \frac{W(650)(6.9\times10^{-8})}{2(1.25\times10^{-4})}\cdot(5-0.65)^2 = 3.39W$$

Then

$$W = 11.8\mu m$$

- **The current capability of a MOSFET is directly proportional to the channel width W. the current gandling capability can be increased by increasing W.**

Example 10.8

① **Determine the inversion carrier mobility from experimental results.**

- **Assume the following transistor parameter : W=15*μm*, L=2*μm*,**
- **C_{ox}=6.9× 10^{-8}*F/cm*^{2}, V_{DS}=0.10V**
- **I_D=35*μA* at V_{GS}=1.5V and I_D=75*μA* at V_{GS}=2.5V**

From Equation (10.68), we can write

$$I_{D2} - I_{D1} = \frac{W\mu_n C_{ox}}{L}(V_{GS2} - V_{GS1})V_{DS}$$

so that

$$75\times10^{-6} - 35\times10^{-6} = \left(\frac{15}{2}\right)\mu_n(6.9\times10^{-8})(2.5-1.5)(0.10)$$

which yields

$$\mu_n = 773 cm^2/V-s$$

We can then determine

$$V_T = 0.625V$$

- **The mobility of carriers in the inversion layer is less than that in the bulk semiconductor due to the surface scattering effect. We will discuss this effect in the next chapter.**

10.3.2 Current-Voltage Relationship-concepts

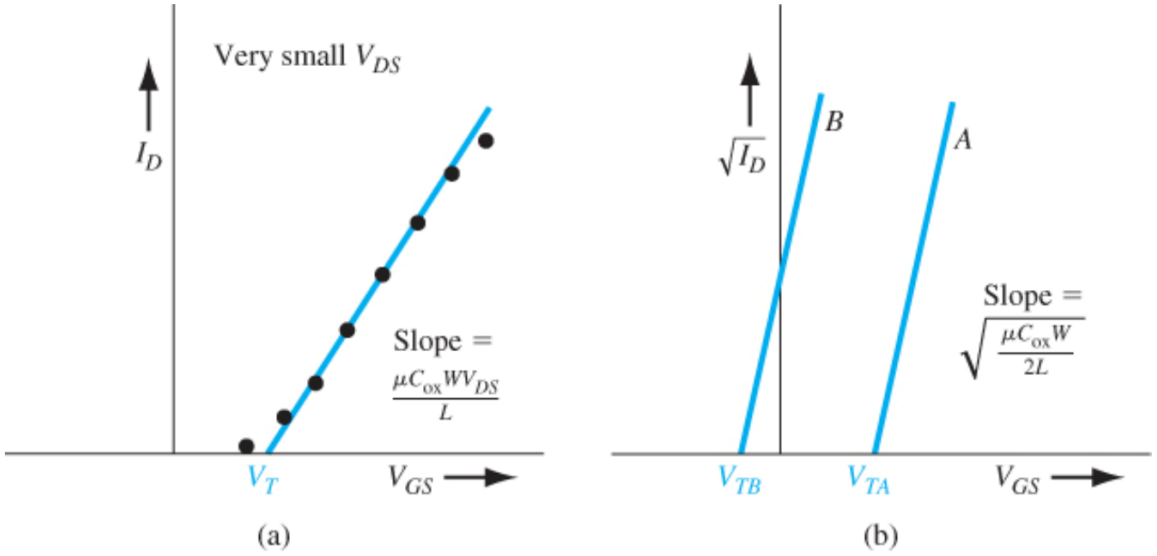


Figure 10.48 | (a) I_D versus V_{GS} (for small V_{DS}) for enhancement mode MOSFET. (b) Ideal $\sqrt{I_D}$ versus V_{GS} in saturation region for enhancement mode (curve A) and depletion mode (curve B) n-channel MOSFETs.

$$I_D = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)V_{DS} \quad (10.68)$$

$$\sqrt{I_D(sat)} = \sqrt{\frac{W\mu_n C_{ox}}{2L}}(V_{GS} - V_T) \quad (10.69)$$